

Invariant measures for the stochastic heat equation on $\mathbb{R}^d$ without drift term

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Abstract

This paper deals with the long term behavior of the solution to the nonlinear stochastic heat equation $\partial u / \partial t - \frac{1}{2} \Delta u = b(u) \dot{W}$ for a Lipschitz continuous diffusion coefficient b which is not necessarily bounded. Using the theory of the stochastic integral laid out by John Walsh, we provide conditions which will guarantee the existence of an invariant measure for a broad range of initial conditions which includes L^2_ρ as well as the Dirac delta distribution δ_0 .

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1 Introduction

We consider the following *stochastic heat equation* (SHE):

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2} \Delta u(t, x) = b(x, u(t, x)) \dot{W}(t, x) & x \in \mathbb{R}^d, t > 0, \\ u(0, \cdot) = \mu(\cdot), \end{cases} \quad (1.1) \quad \boxed{\text{E:SHE}}$$

which can be interpreted as the following stochastic integral equation:

$$u(t, x) = \int_{\mathbb{R}^d} G(t, x - y) \mu(dy) + \int_0^t \int_{\mathbb{R}^d} b(y, u(s, y)) G(t - s, x - y) W(ds, dy). \quad (1.2) \quad \boxed{\text{E:gmsol}}$$

In (1.2), $G(t, x) = (2\pi t)^{-d/2} \exp\left(-\frac{|x|^2}{2t}\right)$ is the heat kernel and the stochastic integral is the *Walsh integral* (see [Wal86; Dal+09]). For some constants $L_b > 0$ and $L_0 \geq 0$, the function $b(x, u)$ satisfies the following Lipschitz and boundedness conditions:

$$|b(x, u) - b(x, v)| < L_b |u - v| \quad \text{and} \quad |b(x, 0)| \leq L_0 \quad \text{for all } u, v \in \mathbb{R}, x \in \mathbb{R}^d. \quad (1.3) \quad \boxed{\text{E:lipcon}}$$

In particular, our assumption allows the linear case $b(x, u) = \lambda u$, which is usually referred to the *parabolic Anderson model* [CM94]. Any $L^2(\Omega)$ -continuous processes, $u = (u(t, x) : t \geq 0, x \in \mathbb{R}^d)$, that is measurable with respect to the natural filtration generated by W that solves (1.2) is referred to as a *mild solution*. See [CH19] and for more.

The noise, $\dot{W}(t, s)$, is a centered Gaussian noise that is white in time and homogeneously correlated in space. Its covariance structure, J , is defined on $C_c^\infty(\mathbb{R}^{d+1}) \times C_c^\infty(\mathbb{R}^{d+1})$ as follows:

$$J(\psi, \phi) := \mathbb{E} [\dot{W}(\psi) \dot{W}(\phi)] = \int_0^\infty ds \int_{\mathbb{R}^d} \Gamma(dx) (\psi(s, \cdot) * \tilde{\phi}(s, \cdot))(x), \quad (1.4) \quad \boxed{\text{E:Cor}}$$

where $\tilde{\phi}(x) := \phi(-x)$ and Γ is a tempered measure on $\mathbb{R}^d$ that is nonnegative and nonnegative definite and is referred to as the *correlation measure*. When it is absolutely continuous with respect to the Lebesgue measure, i.e., $\Gamma(dx) = f(x)dx$, we call f the *correlation function*. The Fourier transform of Γ (in the generalized sense) is also a nonnegative and nonnegative definite tempered measure, which is called *spectral measure* and is denoted by $\hat{f}(d\xi)$. Moreover, in the case where Γ has density f , then $\hat{f}(d\xi) = \hat{f}(\xi)d\xi$.

The aim of this paper is to investigate the conditions required to guarantee the existence of invariant measures for the solution to (1.1), which is a crucial step towards the study of the ergodicity of the system. Existence in unbounded domains has been studied in few papers such as [AM03; MSY20; MSY16; TZ98]. Since the spatial domain is unbounded, one needs to introduce a weight function ρ and the corresponding weighted space $L_\rho^2(\mathbb{R}^d)$ as usual; see Section 2 below for more details. Following the standard procedure as laid out by Da Prato and Zabczyk [DGZ92], the main effort is to show that

$$\sup_{t>0} \mathbb{E} \left(\|u(t, \cdot)\|_{L_\rho^2(\mathbb{R}^d)}^2 \right) < \infty. \quad (1.5) \quad \boxed{\text{E:BddMnt}}$$

In the setting of equation (1.1), Tessitore and Zabczyk [TZ98] verified (1.5) when $\mu(dx) = dx$ under the assumption that the spectral density $\hat{f}$ satisfies ...reference example below.... We will expand this result and show the existence of an invariant measure when $\mu(\cdot)$ satisfies (1.18) and (1.20) below, as well as one of the following:

1. $\mu(dx)$ is a deterministic, locally finite, regular, signed Borel measure with the corresponding *Hahn decomposition* $\mu = \mu_+ - \mu_-$
2. $\mu(dx) = \varphi(x)dx$ where φ is a processes in $L^2(\Omega, L_\rho^2(\mathbb{R}^d))$, namely $\mathbb{E} \left(\|\varphi(\cdot)\|_{L_\rho^2(\mathbb{R}^d)}^2 \right) < \infty$.

Some applicable deterministic initial measures include (see Example 1.7 below):

Dirac delta measure $\delta_0(dx)$, Lebesgue measure dx , $\varphi(x)dx$ with $\varphi \in C^\infty(\mathbb{R}^d)$.

Notation 1.1. We will use $\|\cdot\|_\rho$ instead of $\|\cdot\|_{L_\rho^2(\mathbb{R}^d)}$. Also by defining $\|X\|_p := (\mathbb{E}(|X|^p))^{1/p}$ where $p \geq 1$,

$$J_0(t, x) := (G(t, \cdot) * \mu)(x) = \int_{\mathbb{R}^d} G(t, x - y) \mu(dy) \quad \text{and} \quad J_p(t, x) := (G(t, \cdot) * \|\mu\|_p)(x), \quad (1.6) \quad \boxed{\text{E:J}}$$

where we use the following convention:

$$\|\mu(dx)\|_p = \begin{cases} |\mu|(dx) & \text{when } \mu \text{ is deterministic} \\ \|\varphi(x)\|_p dx & \text{when } \mu \text{ is random.} \end{cases}$$

We also denote

$$I(t, x) := \int_0^t \int_{\mathbb{R}^d} b(y, u(s, y)) G(t - s, x - y) W(ds, dy).$$

We now state Assumption 1.2, under which one can find a unique mild solution to (1.1) with sharp moment estimates; see [CK19; CH19; CD15].

A:General

Assumption 1.2. (i) The initial measure μ must satisfy the following integrability condition:

$$\int_{\mathbb{R}^d} \exp(-a|x|^2) \|\mu(dx)\|_2 < \infty \quad \text{for all } a > 0. \quad (1.7)$$

A:icon

(ii) The spectral measure $\widehat{f}$ satisfies *Dalang's condition*, namely,

$$\Upsilon(\beta) := (2\pi)^{-d} \int_{\mathbb{R}^d} \frac{\widehat{f}(d\xi)}{\beta + |\xi|^2} < \infty \quad \text{for some and therefore any } \beta > 0. \quad (1.8)$$

A:dalcon

Moreover, this solution is usually *intermittent*, namely, the probability moments to (1.1) have certain exponential growth in t . In order to have moments bounded in time as required in (1.5), one has to first identify the exact conditions, under which the moments to (1.1) do not possess exponential growth. This question has been answered in [CK19], where necessary and sufficient conditions are given. In [AM03; MSY20; MSY16], a drift term $f(x, u)$ is included in (1.1) in a crucial way to help to cancel the otherwise exponential growing moments. The absence of such a drift term in this paper makes the problem more challenging.

According to [CK19, Theorem 1.3], in order to have (1.5), one has to have L_b (the Lyapunov exponent of b) in (1.3) small enough, the spatial dimension $d \geq 3$, and we must strengthen (1.8) to the following:

$$\Upsilon(0) < \infty. \quad (1.9)$$

E:UOfinite

R:Cor

Remark 1.3. The assumption on the correlation function in [TZ98] is given through the spectral density $\widehat{f}$ as follows:

$$\widehat{f} \in L^p(\mathbb{R}^d) \quad \text{where } \frac{d-2}{d} < \frac{1}{p}; \quad (1.10)$$

E:CorTZ

see Hypothesis 2.1 (*ibid.*). However as was illustrated in [TZ98, Theorem 3.3], (1.10) was not enough of an assumption to give a concrete example of the existence of an invariant measure and additional assumptions were needed. For example, one needs $d \geq 3$ and the Lipschitz constant of b to satisfy $l^{-2} > 1$, where

$$\tilde{\Gamma} := \left| \sqrt{\mathcal{F}(f)} \right| * \left| \sqrt{\mathcal{F}(f)} \right| \quad \text{and} \quad 1 := \frac{\Gamma(d/2-1)}{4\pi^{d/2}} \int_{\mathbb{R}^d} \tilde{\Gamma}(d\zeta) |\zeta|^{2-d}, \quad (1.11)$$

E:TZspec

where Γ is the Gamma function. These additional assumptions parallel our assumptions (1.9) and (1.15), however one can see that both (1.9) and (1.15) are more compact and easier to check than (1.11).

Notation 1.4. We need to introduce some more notation. Denote

$$\mu_q^* := 1 + \|\mu\|_q \quad \text{and} \quad J_q^*(r, z) = (G(r, \cdot) * \mu_q^*)(x). \quad (1.12)$$

E:qJ*

We remind the reader that for deterministic initial measures

$$\mu_q^* = \mu^* = 1 + |\mu| \quad \text{and} \quad J_q^*(r, z) = J^*(r, z) = (G(r, \cdot) * \mu^*)(x). \quad (1.13)$$

E:J*

Again, whether μ is random or deterministic, we will always use the notation given in (1.12).

We may now a preliminary result used to prove Theorem (1.6) below.

T:Lrho**Theorem 1.5.** *Assume that*

- (i) $\rho : \mathbb{R}^d \rightarrow \mathbb{R}_+$ is an nonnegative $L^1(\mathbb{R}^d)$ function;
- (ii) the initial measure μ , either random or deterministic, satisfies that

$$\mathcal{G}_{\rho,2}(t) = \mathcal{G}_{\rho,\mu_2^*}(t) := \int_{\mathbb{R}^d} (J_2^*(t,x))^2 \rho(x) dx < \infty \quad \text{for all } t > 0, \quad (1.14) \quad \text{E:G}$$

- (iii) the correlation measure Γ satisfies (1.9),
- (iv) the Lipschitz constant L_b of b (see Theorem 2.5 below) satisfies

$$2L_b^2 < \frac{1}{128\Upsilon(0)}. \quad (1.15) \quad \text{E:LipLb}$$

Then for all $t > 0$, $u(t, \cdot) \in L_\rho^2(\mathbb{R}^d)$ a.s. Moreover, for some constant $C > 0$ which does not depend on t , it holds that

$$\mathbb{E} \left(\|u(t, \cdot)\|_\rho^2 \right) \leq C \mathcal{G}_{\rho,2}(t) < \infty. \quad (1.16) \quad \text{E:uInLrho}$$

Proof. The constant C may change from line to line. Under conditions (iii) and (iv), we can apply Fubini's Theorem and the moment bound (2.7) to see that there exists a constant C independent of t such that,

$$\begin{aligned} \mathbb{E} \left(\|u(t, \cdot)\|_\rho^2 \right) &= \mathbb{E} \left(\int_{\mathbb{R}^d} |u(t, x)|^2 \rho(x) dx \right) \\ &\leq C \int_{\mathbb{R}^d} \left(1 + (G(t, \cdot) * \|\mu\|_2)(x) \right)^2 \rho(x) dx \\ &= C \int_{\mathbb{R}^d} \left((G(t, \cdot) * (1 + \|\mu\|_2))(x) \right)^2 \rho(x) dx \\ &= C \mathcal{G}_{\rho,2}(t) < \infty \end{aligned}$$

This proves Theorem 1.5. □

Here is the main theorem of the paper:

T:Main

Theorem 1.6. *Assume that all assumptions of Theorem 1.5 hold. Let $q > 2$ be arbitrary but fixed. Moreover, suppose that*

- (i) if ρ is an admissible weight and there exists another admissible weight $\hat{\rho}$ such that

$$\int_{\mathbb{R}^d} \frac{\rho(x)}{\hat{\rho}(x)} dx < \infty, \quad (1.17) \quad \text{E:rho rho}$$

- (ii) the weight function $\hat{\rho}$ and the initial condition satisfy the following condition:

$$\limsup_{t>0} \mathcal{G}_{\hat{\rho},q}(t) < \infty, \quad (1.18) \quad \text{E:InitData}$$

- (iii) for some $\alpha \in (0, 1/2)$, the spectral measure $\hat{f}$ satisfies that

$$\mathcal{H}_\alpha(t) := \int_0^t dr r^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp(-r|\xi|^2) < \infty \quad \text{for all } t > 0, \quad (1.19) \quad \text{E:H}$$

(iv) for $\alpha \in (q^{-1}, 2^{-1})$,

$$\int_0^t [\mathcal{G}_{\hat{\rho},q}(s) \mathcal{H}_\alpha(s)]^{q/2} ds < \infty \quad \text{for all } t > 0, \quad (1.20) \quad \boxed{\text{E:GH-int}}$$

(v) the condition (1.15) is strengthened to

$$qL_b^2 < \frac{1}{128\Upsilon(0)}.$$

Then we have that

(a) for any $t_0 > 0$, the sequence of laws of $\{\mathcal{L}u(t, \cdot)\}_{t \geq t_0}$ is tight, i.e., for any $\epsilon \in (0, 1)$, there exists a compact set $\mathcal{K} \subset L_\rho^2(\mathbb{R}^d)$ such that

$$\mathcal{L}u(t, \cdot)(\mathcal{K}) := \mathbb{P}(u(t, \cdot) \in \mathcal{K}) \geq 1 - \epsilon, \quad \text{for all } t \geq t_0; \quad (1.21) \quad \boxed{\text{E:cpt}}$$

(b) there exists an invariant measure for the laws $\{\mathcal{L}u(t, \cdot)\}_{t > 0}$.

E:Gamma+f

Example 1.7 (Correlation measures). Here are a few examples of correlation measures Γ that satisfy condition (1.20). The Bessel potential (see, e.g., Section 1.2.2 of [Gra14])... .. TBD ...

E:Init

Example 1.8 (Initial conditions). Here are few examples:

1. If the initial condition μ is the Lebesgue measure, than the left-hand side of (1.18) is equal to $\|\widehat{\rho}\|_{L^1(\mathbb{R}^d)}$, which is finite and does not depend on t .
2. If $\mu = \delta_0$,

Proof of Theorem 1.6 (b). Fix an arbitrary $T_0 > 0$ and denote

$$U(T) := \frac{1}{T} \int_{T_0}^{T+T_0} \mathcal{L}(u(t, \cdot)) dt, \quad T > 0.$$

We claim that the family of laws $U(T, \cdot)$ for $T > 0$ is tight in $L_\rho^2(\mathbb{R}^d)$. Indeed, for any $\epsilon \in (0, 1)$, by part (a), there exists a compact set $\mathcal{K} \in L_\rho^2(\mathbb{R}^d)$ such that (1.21) holds. This implies that

$$U(T)(\mathcal{K}^c) = \frac{1}{T} \int_{T_0}^{T+T_0} \mathcal{L}(u(t, \cdot))(\mathcal{K}^c) dt \leq \frac{1}{T} \int_{T_0}^{T+T_0} \epsilon dt = \epsilon, \quad \text{for all } T > 0.$$

Let $\{T_n\}_{n \in \mathbb{N}}$ be any deterministic sequence such that $T_n \uparrow \infty$. Since $\{U(T_n)\}_{n \geq 1}$ is a tight sequence of measures, there exists a subsequence $\{U(T_{n_m})\}_{m \geq 1}$ that converges weakly to a measure, say, η , on $L_\rho^2(\mathbb{R}^d)$. Then one can apply for the *Krylov-Bogoliubov theorem* (see, e.g., [DZ96, Theorem 3.1.1]) to conclude that the measure η is the invariant measure for $\mathcal{L}(u(t, \cdot))$, $t \geq T_0$. Finally, we can choose T_0 to be arbitrarily close to zero. This completes the proof of part (c) of Theorem 1.6. $\square$

2 Preliminaries

S:Prelim

Let ρ be some nonnegative function on $\mathbb{R}^d$. We denote by $L_\rho^2(\mathbb{R}^d)$ the weighted Hilbert space with the inner produce $\langle f, g \rangle_\rho := \int_{\mathbb{R}^d} f(x)g(x)\rho(x) dx$ and set $\|f\|_\rho^2 := \int_{\mathbb{R}^d} f(x)^2\rho(x) dx$. Following Tessitore and Zabczyk [TZ98], we make the following definition:

D:Rho

Definition 2.1. $\rho : \mathbb{R}^d \mapsto \mathbb{R}$ is called an *admissible weight* if ρ is a strictly positive, bounded and continuous function in $L^1(\mathbb{R}^d)$ such that for $T > 0$ there exists a constant $C_\rho(T)$ such that

$$(G(t, \cdot) * \rho(\cdot))(x) \leq C_\rho(T)\rho(x) \quad \text{for all } t \in [0, T] \text{ and } x \in \mathbb{R}^d, \quad (2.1)$$

E:aw

or equivalently,

$$\sup_{x \in \mathbb{R}^d, t \in [0, T]} \frac{(G(t, \cdot) * \rho(\cdot))(x)}{\rho(x)} \leq C_\rho(T). \quad (2.2)$$

E:awbd1

Several examples of admissible weights are (see Example in Section 2 of [TZ98])

$$\begin{aligned} \rho(x) &= \exp(-a|x|), & a > 0 \quad \text{and} \\ \rho(x) &= (1 + |x|^a)^{-1}, & a > d. \end{aligned} \quad (2.3)$$

E:admRho

R:admRho

Remark 2.2. The smaller the weight function $\rho(\cdot)$ (not necessarily to be admissible here) is, the larger the space $L_\rho^2(\mathbb{R}^d)$ is. For example, one may choose ρ to be either a nonnegative function with compact support or the heat kernel itself $G(1, \cdot)$. In both cases, ρ is smaller than those in (2.3) (up to a constant). However, one can easily check that the admissible condition (2.1) excludes these two cases. It is much less obvious that the following weight function is not admissible:

$$\rho(x) = \exp(-a|x|^b), \quad x \in \mathbb{R}^d, \quad \text{with } a > 0 \text{ and } b \in (1, 2) \text{ fixed.}$$

P:weight

Proposition 2.3 (Proposition 2.1 of [TZ98]). *For any admissible weight ρ , the operators on $L_\rho^2(\mathbb{R}^d)$ defined by $\varphi \mapsto (G(t, \cdot) * \varphi(\cdot))(x)$ can be extended to a C_0 -semigroup on L_ρ^2 . Moreover, if $\widehat{\rho}$ is another admissible weight such that*

$$\int_{\mathbb{R}^d} \frac{\rho(x)}{\widehat{\rho}(x)} dx < \infty,$$

then for any $t < 0$, the operators defined above are compact from $L_\rho^2(\mathbb{R}^d)$ to $L_{\widehat{\rho}}^2(\mathbb{R}^d)$.

We need Lemma 2.4 to how that $\varphi \in L_\rho^2$ will satisfy (1.7).

L:RhoDom

Lemma 2.4. *For any admissible weight function ρ and $a > 0$, there exists a constant $K_{\rho,a} > 0$ such that*

$$\exp(-a|x|^2) \leq K_{\rho,a} \rho(x), \quad \text{for all } x \in \mathbb{R}^d. \quad (2.4)$$

E:RhoDom

Proof. Let ρ be an arbitrary admissible weight function and $a > 0$. By (2.1),

$$\begin{aligned} \frac{\exp(-a|x|^2)}{\rho(x)} &= \frac{\exp(-a|x|^2)}{\rho(x)} \frac{(G(1, \cdot) * \rho(\cdot))(x)}{(G(1, \cdot) * \rho(\cdot))(x)} \\ &\leq \left(\sup_{x \in \mathbb{R}^d, t \in [0, 1]} \frac{(G(t, \cdot) * \rho(\cdot))(x)}{\rho(x)} \right) \frac{\exp(-a|x|^2)}{(G(1, \cdot) * \rho(\cdot))(x)} \\ &\leq C_\rho(1) \frac{\exp(-a|x|^2)}{(G(1, \cdot) * \rho(\cdot))(x)}. \end{aligned}$$

Noticing that for some constant $\Theta > 0$, $\rho(x) \geq \Theta \prod_{\ell=1}^d \mathbb{I}_{[-1, 1]}(x_\ell)$, a simple exercise shows that

$$C_{\rho,a} := \sup_{x \in \mathbb{R}^d} \frac{\exp(-a|x|^2)}{(G(1, \cdot) * \rho(\cdot))(x)} \leq \frac{1}{\Theta} \left(\sup_{x \in \mathbb{R}} \frac{\exp(-ax^2)}{(G(1, \cdot) * \mathbb{I}_{[-1, 1]}(\cdot))(x)} \right)^d < \infty.$$

This proves the lemma with $K_{\rho,a} = C_\rho(1)C_{\rho,a}$. $\square$

T:H(t)Bdd

Theorem 2.5 (Theorem 1.2 [CH19]). *Under Assumption 1.2, the SPDE (1.1) has a unique random field solution starting from μ and the solution is $L^2(\Omega)$ continuous.*

T:ExUn

Theorem 2.6 (Theorem 1.7 [CH19]). *Under Assumption 1.2, for any $t > 0$ and $x \in \mathbb{R}^d$, the solution to (1.1), $u(t, x)$, given by (1.2) is in $L^p(\Omega)$ and for $p \geq 2$*

$$\|u(t, x)\|_p \leq [\bar{\varsigma} + \sqrt{2}(G(t, \cdot) * \|\mu\|_p)(x)] H(t; \gamma_p)^{1/2}, \quad (2.5)$$

E:mb

where $\bar{\varsigma} = L_0/L_b$, $\gamma_p = 32pL_b^2$ and the function $H(t; \gamma_p)$ is nondecreasing in t (see [CH19] for the concrete definition of H). Moreover, if we assume that

$$\Upsilon(0) < \infty \quad \text{and} \quad \gamma_p < \frac{1}{4\Upsilon(0)}, \quad (2.6)$$

A:dalcon0

then this function $H(t; \gamma_p)$ is bounded in t . Hence, by choosing $C = (\sqrt{2} \vee \bar{\varsigma}) \sup_{t \geq 0} H(t; \gamma_p)^{1/2} < \infty$, we have that

$$\|u(t, x)\|_p \leq C(L_0 + J_p(t, x)), \quad (2.7)$$

E:mb0

where L_0 is from (1.3).

The following factorization property of the heat kernel will be used in this paper:

$$G(s - r, y - z)G(r, z - \sigma) = G(s, y - \sigma)G\left(\frac{r(s - r)}{s}, z - \sigma - \frac{r}{s}(y - \sigma)\right). \quad (2.8)$$

E:gsm

3 Proof of part (a) of Theorem 1.6

S:mainproof

For $\alpha \in (0, 1)$, $t > 0$ and $x \in \mathbb{R}^d$, define formally

$$\begin{aligned} (F_\alpha f)(t, x) &:= \int_0^t \int_{\mathbb{R}^d} (t - s)^{\alpha-1} G(t - s, x - y) f(s, y) \, ds dy \quad \text{and} \\ (Y_\alpha f)(t, x) &:= \int_0^t \int_{\mathbb{R}^d} (t - s)^{-\alpha} G(t - s, x - y) f(s, y) W(ds, dy). \end{aligned} \quad (3.1)$$

E:Ops

For F_α , the Step 1 of the proof of [TZ98, Theorem 3.1] showed the following proposition:

P:F-Comp

Proposition 3.1. *Let ρ and $\hat{\rho}$ be given in condition (i) of Theorem 1.6 (see (1.17)). For any $q > 2$, $t_0 > 0$ and $\alpha \in (q^{-1}, 2^{-1})$, the operator F_α as an operator from $L^q((0, t_0); L_\rho^2)$ to $L_\rho^2(\mathbb{R}^d)$ is compact.*

As for Y_α , we have the following result:

L:yineq

Lemma 3.2. *For any $\alpha \in (0, 1/2)$, denote (with slightly abuse of notation)*

$$Y(s, y) := [Y_\alpha b(\circ, u(\cdot, \circ))](s, y) = \int_0^s \int_{\mathbb{R}^d} (s - r)^{-\alpha} G(s - r, y - z) b(z, u(r, z)) W(dr, dz). \quad (3.2)$$

E:Y

Under the assumptions of Theorem 1.5 and (iii) of Theorem 1.6, then for all $q \geq 2$, $s > 0$ and $y \in \mathbb{R}^d$,

$$\|Y(s, y)\|_q^2 \leq C_q [J_q^*(s, y)]^2 \mathcal{H}_\alpha(s). \quad (3.3)$$

E:NormY

Moreover, if

(iv') for some $q \geq 2$ and $\alpha \in (0, 1/2)$, the weight function ρ (not necessarily admissible)

$$\int_0^t [\mathcal{G}_{\rho,q}(s) \mathcal{H}_\alpha(s)]^{q/2} ds < \infty, \quad \text{for all } t > 0, \quad (3.4) \quad \boxed{\text{E:GH-int'}}$$

then there exists a constant $\Theta = \Theta(q, L_b, L_0, \alpha)$ such that

$$\mathbb{E} \left(\int_0^t \|Y(s, \cdot)\|_\rho^q ds \right) \leq \Theta \int_0^t [\mathcal{G}_{\rho,q}(s) \mathcal{H}_\alpha(s)]^{q/2} ds < \infty. \quad (3.5) \quad \boxed{\text{E:yineq}}$$

Proof. In the proof, we use C to denote a generic constant that may change its value at each appearance. First, an application of Minkowski's inequality shows that

$$\mathbb{E} \left(\|Y(s, \cdot)\|_\rho^q \right) = \left\| \int_{\mathbb{R}^d} Y(s, y)^2 \rho(y) dy \right\|_{q/2}^{q/2} \leq \left(\int_{\mathbb{R}^d} \|Y(s, y)\|_q^2 \rho(y) dy \right)^{q/2}. \quad (3.6) \quad \boxed{\text{E:EYqr}}$$

Now we look to find an appropriate upper bound for $\|Y(s, y)\|_q^2$. By the Burkholder-Davis-Gundy inequality, as well as Minkowski's integral inequality to see that

$$\begin{aligned} \|Y(s, y)\|_q^2 &\leq C_q \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1) \|b(z_1, u(r, z_1))\|_q f(z_1-z_2) \\ &\quad \times G(s-r, y-z_2) \|b(z_2, u(r, z_2))\|_q. \end{aligned}$$

Note that for the Lipschitz condition in (1.3), we have that

$$|b(x, u)| \leq |b(x, u) - b(x, 0)| + |b(x, 0)| \leq L_b |u| + L_0 \leq C(1 + |u|), \quad C := L_b \vee L_0.$$

We apply this and the moment bound (2.7) to $\|b(z_i, u(r, z_i))\|_q$ above to see that

$$\|b(z_i, u(r, z_i))\|_q \leq C \left(1 + \|u(r, z_i)\|_q \right) \leq C \left(1 + (G(r, \cdot) * \|\mu\|_q)(z_i) \right) = C J_q^*(r, z_i), \quad i = 1, 2, \quad (3.7) \quad \boxed{\text{E:NormB}}$$

where J_q^* is defined in (1.12). Therefore,

$$\begin{aligned} \|Y(s, y)\|_q^2 &\leq C \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 f(z_1-z_2) \prod_{i=1}^2 \left(G(s-r, y-z_i) J_q^*(r, z_i) \right) \\ &= C \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} \mu_q^*(d\sigma_1) \mu_q^*(d\sigma_2) \iint_{\mathbb{R}^{2d}} dz_1 dz_2 \\ &\quad \times f(z_1-z_2) \prod_{i=1}^2 \left(G(s-r, y-z_i) G(r, z_i-\sigma_i) \right) \\ &= C \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} \mu_q^*(d\sigma_1) \mu_q^*(d\sigma_2) G(s, y-\sigma_1) G(s, y-\sigma_2) \\ &\quad \times \iint_{\mathbb{R}^{2d}} dz_1 dz_2 f(z_1-z_2) \prod_{i=1}^2 G\left(\frac{r(s-r)}{s}, z_i-\sigma_i - \frac{r}{s}(y-\sigma_i) \right) \\ &\leq C(2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} \mu_q^*(d\sigma_1) \mu_q^*(d\sigma_2) G(s, y-\sigma_1) G(s, y-\sigma_2) \\ &\quad \times \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2 \right), \end{aligned}$$

where we have applied (2.8) and Plancherel's theorem. Therefore,

$$\|Y(s, y)\|_q^2 \leq C(2\pi)^{-2d} [J_q^*(s, y)]^2 \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right). \quad (3.8) \quad \boxed{\text{E_ : NormY}}$$

By symmetry of $r(s-r)/s$ term and the fact that $r(s-r)/s \geq r/2$ for all $r \in [0, s/2]$, we see that the above double integral is bounded by

$$\begin{aligned} &\leq 2 \int_0^{s/2} dr r^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{r}{2} |\xi|^2\right) \\ &= 2^{2(1-\alpha)} \int_0^{s/4} dr r^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp(-r |\xi|^2) \\ &\leq 2^{2(1-\alpha)} \int_0^s dr r^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp(-r |\xi|^2) \\ &= 2^{2(1-\alpha)} \mathcal{H}_\alpha(s). \end{aligned}$$

Plugging the above bound back to (3.8) proves (3.3).

It remains to prove (3.5). By the definition of $\mathcal{G}_\rho(t)$ in (1.14) and by (3.3), we see that

$$\int_{\mathbb{R}^d} \|Y(s, y)\|_q^2 \rho(y) dy \leq C \mathcal{G}_{\rho, q}(s) \mathcal{H}_\alpha(s).$$

Plugging the above expression to the far right-hand of (3.6) proves (3.5). Finally, the finiteness of the upper bound in (3.5) is guaranteed by condition (3.4). This completes the proof of Lemma 3.2. $\square$

L:Factorize

Lemma 3.3 (Factorization lemma). *Let $u(t, x)$ be the solution to (1.1) and Y be given in (3.2). Under the conditions of Theorem 1.5 and (iii) of Theorem 1.6, for all $\alpha \in (0, 1/2)$, the following factorization holds*

$$\frac{\sin(\alpha\pi)}{\pi} \int_0^t (t-s)^{\alpha-1} [G(t-s, \cdot) * Y(s, \cdot)](x) ds = \int_0^t \int_{\mathbb{R}^d} G(t-r, x-z) b(z, u(r, z)) W(dr, dz).$$

As a consequence,

$$u(t, x) = J_0(t, x) + \frac{\sin(\alpha\pi)}{\pi} [F_\alpha Y](t, x), \quad \text{for all } t > 0 \text{ and } x \in \mathbb{R}^d. \quad (3.9) \quad \boxed{\text{E:1fac}}$$

Proof. The lemma is straightforward provided that one can switch the orders of stochastic and ordinary integrals:

$$\begin{aligned} &\int_0^t (t-s)^{\alpha-1} [G(t-s, \cdot) * Y(s, \cdot)](x) ds \\ &= \int_0^t ds (t-s)^{\alpha-1} \int_{\mathbb{R}^d} dy G(t-s, x-y) \\ &\quad \times \int_0^s \int_{\mathbb{R}^d} (s-r)^{-\alpha} G(s-r, y-z) b(z, u(r, z)) W(dr, dz) \\ &= \int_0^t ds (t-s)^{\alpha-1} \int_0^s \int_{\mathbb{R}^d} (s-r)^{-\alpha} G(t-r, x-z) b(z, u(r, z)) W(dr, dz) \quad (3.10) \quad \boxed{\text{E:Fubini}_1} \\ &= \int_0^t \int_{\mathbb{R}^d} W(dr, dz) G(t-r, x-z) b(u(r, z)) \int_r^t ds (s-r)^{-\alpha} (t-s)^{\alpha-1} \quad (3.11) \quad \boxed{\text{E:Fubini}_2} \end{aligned}$$

$$= \frac{\pi}{\sin(\alpha\pi)} \int_0^t \int_{\mathbb{R}^d} G(t-r, x-z) b(u(r, z)) W(dr, dz),$$

where the last step is the Beta integral which requires that $\alpha \in (0, 1)$. It remains to justify the two applications of the stochastic Fubini's theorem (see Theorem 5.30 of Chapter one in [Dal+09], or also [Wal86] or Theorem 4.18 of [DZ92]) in (3.10) and (3.11) in the following two steps.

Step 1. In this step, we justify the change of orders in (3.10). It suffices to prove the following condition:

$$\begin{aligned} I_1 &:= \int_{\mathbb{R}^d} dy G(t-s, x-y) \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 \\ &\quad \times f(z_1 - z_2) \left(\prod_{i=1}^2 G(s-r, y-z_i) \right) \mathbb{E} \left(\prod_{i=1}^2 b(z_i, u(r, z_i)) \right) \\ &= \int_{\mathbb{R}^d} dy G(t-s, x-y) \|Y(s, y)\|_2^2 \\ &< +\infty. \end{aligned}$$

But this follows immediately from (3.3). Indeed

$$\begin{aligned} \int_{\mathbb{R}^d} dy G(t-s, x-y) \|Y(s, y)\|_2^2 &\leq C \int_{\mathbb{R}^d} dy G(t-s, x-y) J_2^*(s, y) \mathcal{H}_\alpha(s) \\ &= C \mathcal{H}_\alpha(s) \int_{\mathbb{R}^d} dy G(t-s, x-y) \int_{\mathbb{R}^d} \mu_2^*(dx) G(s, y-z) \\ &= C \mathcal{H}_\alpha(s) J_2^*(t, x). \end{aligned}$$

Then Condition (1.19) implies that $I_1 < \infty$.

A type has been made above when using (3.3) to bound $\|Y(s, y)\|_2^2$. Note that we need to show

$$\begin{aligned} \int_{\mathbb{R}^d} dy G(t-s, x-y) \|Y(s, y)\|_2^2 &\leq C \int_{\mathbb{R}^d} dy G(t-s, x-y) (J_2^*(s, y))^2 \mathcal{H}_\alpha(s) \\ &= \mathcal{H}_\alpha(s) \int_{\mathbb{R}^d} dy G(t-s, x-y) (J_2^*(s, y))^2 < \infty \end{aligned}$$

However, because s and t are fixed, this will follow from the assumption (1.14) which says that $\mathcal{G}_{\rho,2}(s) < \infty$ for each $s > 0$. This is due to the fact that $\exp(-|x|^2) \leq C\rho(x)$. **Step**

2. Similarly, as for (3.11), we need to show that

$$\begin{aligned} I_2 &:= \int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 \\ &\quad \times f(z_1 - z_2) \left(\prod_{i=1}^2 G(t-r, x-z_i) \right) \mathbb{E} \left(\prod_{i=1}^2 b(z_i, u(r, z_i)) \right) < \infty. \end{aligned}$$

By the Cauchy Schwartz inequality, (3.7) and because $\alpha \in (0, 1/2)$,

$$I_2 \leq C \int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 f(z_1 - z_2) \prod_{i=1}^2 \left(G(t-r, x-z_i) J_2^*(r, z_i) \right)$$

$$= C' \int_0^t dr (t-r)^{-\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 f(z_1 - z_2) \prod_{i=1}^2 \left(G(t-r, x - z_i) J_2^*(r, z_i) \right).$$

Now by the same arguments as those leading to (3.3) (with 2α there replaced by α), we see that

$$I_2 \leq C J_*^2(t, x) \int_0^t dr r^{-\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{r(t-r)}{r} |\xi|^2\right),$$

which is finite by (1.19) where we replace α with $\alpha/2$. Indeed, since $r(t-r) \geq tr/2$ for $r \in [0, t]$, then

$$\int_0^t dr r^{-\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{r(t-r)}{r} |\xi|^2\right) \leq \int_0^t dr r^{-\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{r}{2} |\xi|^2\right) < \infty.$$

This completes the proof of Lemma 3.3. $\square$

We are now ready to prove part (b) of Theorem 1.6.

Proof of Theorem 1.6 (b). In this proof we will use $\stackrel{\mathcal{L}}{=}$ to denote equality in law, otherwise, the equality holds almost surely. Let $t_0 > 0$ and consider the following restarted equation:

$$\begin{cases} \frac{\partial v}{\partial t}(t, x) - \frac{1}{2} \Delta v(t, x) = b(x, v(t, x)) \dot{W}_{t_0/2}(t, x) & x \in \mathbb{R}^d, t > 0, \\ v(0, \cdot) = u(t_0/2, \cdot), \end{cases} \quad (3.12)$$

E:SHErandin

where $\dot{W}_{t_0/2}(t, x) = \dot{W}(t + t_0/2, x)$ denotes the time shifted noise. [Should the time shifted noise be equality in law?](#) We note that (1.16) implies that this random initial condition, $u(t_0/2, \cdot)$ satisfies (1.7) and thus the solution to (3.12) exists and we denote the solution as v . By the Markov property of the solution to (1.1), we may say that for any $t \geq 0$, $u(t + t_0/2, x) \stackrel{\mathcal{L}}{=} v(t, x)$, where $\mathcal{L}$ means the equality is in law. In other words, for all $t \geq 0$, the following holds:

$$u(t + t_0/2, x) \stackrel{\mathcal{L}}{=} v(t, x) = (G(t, \cdot) * u(t_0/2, \cdot))(x) + \int_0^t \int_{\mathbb{R}^d} G(t-s, x-y) b(y, v(s, y)) W_{t_0/2}(ds, dy).$$

We will use the notation Y_v and $\mathcal{G}_{\hat{\rho}, q}^v$ where

$$Y_v(s, y) := \int_0^s \int_{\mathbb{R}^d} (s-r)^{-\alpha} G(s-r, y-z) b(z, v(r, z)) W_{t_0/2}(dr, dz),$$

and

$$\begin{aligned} \mathcal{G}_{\hat{\rho}, q}^v(s) &= \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} G(s, x-y) \|v(0, y)\|_q dy \right)^2 \hat{\rho}(x) dx \\ &= \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} G(s, x-y) \|u(t_0/2, y)\|_q dy \right)^2 \hat{\rho}(x) dx. \end{aligned}$$

We define $\mathcal{G}_{\hat{\rho}, q}^u(s)$ and Y_u in an equivalent manner. We wish to be able to apply the factorization result (3.9) and also the bound in (3.5) for Y_v . This requires us to show that $v(0, \cdot) = u(t_0/2, \cdot)$ satisfies (1.14) and (3.4) with $\hat{\rho}$. Indeed, Hölder's inequality and the admissible weight property imply that

$$\mathcal{G}_{\hat{\rho}, q}^v(s) = \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} G(s, x-y) \|u(t_0/2, y)\|_q dy \right)^2 \hat{\rho}(x) dx$$

$$\begin{aligned}
&\leq \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} G(s, x-y) \|u(t_0/2, y)\|_q^2 dy \hat{\rho}(x) dx \\
&\leq C(t_0) \int_{\mathbb{R}^d} \|u(t_0/2, y)\|_q^2 \hat{\rho}(y) dy.
\end{aligned}$$

Now by (2.7),

$$\begin{aligned}
C(t_0) \int_{\mathbb{R}^d} \|u(t_0/2, y)\|_q^2 \hat{\rho}(y) dy &\leq C(t_0) \int_{\mathbb{R}^d} \left(G(t_0/2, \cdot) * (1 + \|\mu\|_q) \right)^2(y) \hat{\rho}(y) dy \\
&= C(t_0) \mathcal{G}_{\hat{\rho}, q}^u(t_0/2) < \infty,
\end{aligned}$$

where the finiteness follows from (1.18). Finally, this and the continuity of $\mathcal{H}_\alpha(s)$ for $s \in [0, t_0/2]$ shows that,

$$\int_0^{t_0/2} (\mathcal{G}_{\hat{\rho}, q}^v(s) \mathcal{H}_\alpha(s))^{q/2} ds \leq \mathcal{G}_{\hat{\rho}, q}^u(t_0/2)^{q/2} \int_0^{t_0/2} (\mathcal{H}_\alpha(s))^{q/2} ds < \infty.$$

Thus we are justified in saying that by (3.9),

$$u(t_0, x) \stackrel{\mathcal{L}}{=} v(t_0/2, x) = \left(G(t_0/2, \cdot) * u(t_0/2, \cdot) \right)(x) + \frac{\sin(\alpha\pi)}{\pi} [F_\alpha Y_v](t_0/2, x). \quad (3.13)$$

We now define the set $\mathcal{K}(\Lambda)$ as

$$\mathcal{K}(\Lambda) = \{ (G(t_0, \cdot) * y(\cdot))(x) + (F_\alpha h)(x), \quad \text{where } \|y\|_{\hat{\rho}} \leq \Lambda \text{ and } \|h\|_{L^q(0,1,L_{\hat{\rho}}^2)} \leq \Lambda \}.$$

Then by Proposition 3.1, $\mathcal{K}(\Lambda)$ is relatively compact in $L_{\hat{\rho}}^2$ for $\Lambda > 0$. Moreover, by (3.5) and Chebyshev's inequality we have

$$\begin{aligned}
\mathbb{P}\{u(t_0, \cdot) \notin \mathcal{K}(\Lambda)\} &= \mathbb{P}\{v(t_0/2, \cdot) \notin \mathcal{K}(\Lambda)\} \\
&\leq \mathbb{P} \left[\left(\int_0^{t_0/2} \|Y_v(s, \cdot)\|_{\hat{\rho}}^q ds \right)^{1/q} > \frac{\pi\Lambda}{\sin(\alpha\pi)} \right] + \mathbb{P} \left[\|u(t_0/2, \cdot)\|_{\hat{\rho}} > \Lambda \right] \\
&\leq \frac{\sin^q(\alpha\pi)}{\pi^q \Lambda^q} \mathbb{E} \int_0^{t_0/2} \|Y_v(s, \cdot)\|_{\hat{\rho}}^q ds + \frac{1}{\Lambda^2} \mathbb{E} \left(\|u(t_0/2, \cdot)\|_{\hat{\rho}}^2 \right) \\
&\leq \frac{\sin^q(\alpha\pi)}{\pi^q \Lambda^q} \Theta \int_0^{t_0/2} (\mathcal{G}_{\hat{\rho}, q}^v(s) \mathcal{H}_\alpha(s))^{q/2} ds + \frac{1}{\Lambda^2} \mathbb{E} \left(\|u(t_0/2, \cdot)\|_{\hat{\rho}}^2 \right) < \infty
\end{aligned} \quad (3.14)$$

E:Restartbd

With all this said, (3.14) implies that

$$\mathbb{P}\{u(t_0, \cdot) \in \mathcal{K}(\Lambda)\} \geq 1 - \frac{\sin^q(\alpha\pi)}{\pi^q \Lambda^q} \left(\Theta \int_0^{t_0/2} (\mathcal{G}_{\hat{\rho}, q}^v(s) \mathcal{H}_\alpha(s))^{q/2} ds \right) - \frac{1}{\Lambda^2} \mathbb{E} \left(\|u(t_0/2, \cdot)\|_{\hat{\rho}}^2 \right)$$

Therefore, we can choose $\Lambda > 0$ big enough such that

$$\frac{\sin^q(\alpha\pi)}{\pi^q \Lambda^q} \left(\Theta \int_0^{t_0/2} (\mathcal{G}_{\hat{\rho}, q}^v(s) \mathcal{H}_\alpha(s))^{q/2} ds \right) + \frac{1}{\Lambda^2} \mathbb{E} \left(\|u(t_0/2, \cdot)\|_{\hat{\rho}}^2 \right) < \epsilon,$$

then we get that

$$\mathbb{P}\{u(t_0, \cdot) \in \mathcal{K}(\Lambda)\} \geq 1 - \epsilon.$$

We can now deduce from this and the semigroup property of the solution to the heat equation that

$$\mathbb{P}\{u(t, \cdot) \in \mathcal{K}(\Lambda)\} \geq (1 - \epsilon)\mathbb{P}\{\|u(t - t_0/2, \cdot)\|_{\hat{\rho}} < \Lambda\}.$$

We now apply Chebyshev's inequality and (1.18) to see that for all $t > t_0/2$ that there exists a $C > 0$ such that

$$\mathbb{P}\{\|u(t - t_0/2, \cdot)\|_{\hat{\rho}} \geq \Lambda\} \leq \frac{\mathbb{E}\left(\|u(t - t_0/2, \cdot)\|_{\hat{\rho}}^2\right)}{\Lambda^2} \leq C \frac{\mathcal{G}_{\hat{\rho}, q}\left(t - \frac{t_0}{2}\right)}{\Lambda^2} \leq \frac{C}{\Lambda^2}.$$

Thus, by enlarging Λ if necessary,

$$\mathbb{P}\{\|u(t - t_0/2, \cdot)\|_{\hat{\rho}} \geq \Lambda\} \leq \epsilon, \quad \forall t \geq t_0/2,$$

thus completing the proof of (1.21). □

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